

Towards a First Principles Model of Curling Ice Friction and Curling Stone Dynamics

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ABSTRACT

Scientific investigations to measure and explain the curl (lateral displacement) of a granite stone sliding on ice in the sport of curling go back almost 100 years (Harrington, 1924). Nevertheless, some prominent researchers in the field remain baffled as to the physical explanation of this lateral displacement. And no one has thus far been able to produce a quantitative model, from first principles, that predicts all the documented characteristics of the observed curl. In this paper, we describe our progress towards producing such a numerical model.

After reviewing the history of scientific research on curling, we summarize the experimental observations (quantitative and qualitative) that need to be explained by any successful model; we also summarize the areas of general agreement. Lastly, we formulate a numerical model of ice friction for a non-rotating curling stone, which successfully predicts its longitudinal deceleration. The model also successfully predicts the rotational deceleration of a non-translating curling stone. However, a numerical model for the lateral acceleration that produces the observed curl remains elusive.

KEY WORDS: ice; ice friction; curling

INTRODUCTION

Despite almost 100 years of scientific investigation, the answer to the question, “What puts the curl in a curling rock?” remains elusive. In this paper, we review the history of these scientific investigations and examine some previous explanations of the curl that don’t quite work. We also summarize the experimental observations that will need to be explained by any successful model. And we present our own initial steps towards developing, from first principles, a testable model of ice friction in curling, based in part on our earlier theoretical work on ice friction in the sports of speed skating, bobsleigh and skeleton (Penny et al., 2007; Lozowski and Szilder, 2013; Lozowski et al., 2013; Lozowski et al., 2014a; Lozowski et al., 2014b). In order to make some progress, we have simplified curling dynamics by ignoring everything but two essential elements. First, we examine the frictional interaction between a single pebble and the running band of a curling rock. Second, we derive a friction coefficient based on thermodynamic equilibrium, without considering the details of dry and wet friction.

HISTORY

The Sport of Curling

Curling is a winter team sport played by two teams of four players on a rectangular sheet of ice. It is sometimes called “The Roarin’ Game” because of the rumbling sound made by the 20 kg granite stones as they travel across the pebbled ice. Curling originated in the 16th Century in Scotland, where it was played outdoors on frozen ponds and lochs. Today it is played indoors on carefully prepared ice surfaces.

Rosenberg (2005) and Kietzig et al. (2010) have addressed the physics of ice friction and why it is slippery. But it remains controversial. For example, Makkonen and Tikanmaki (2014) dispute some of the claims of Kietzig et al. (2010).

There is no need to summarize the rules and strategy of the game here. However, the essence of the game is to launch curling stones down the ice, in such a way that they come to a halt close to the centre (the “button”) of a circular target (the “house”) at the opposite end of the ice sheet. To get to the target, the player may need to manoeuvre the rock around other rocks that are in its way. In order to do so, the player imparts a slow rotation to the rock as he/she launches it down the ice. As a result of this rotation, the rock is displaced laterally from its original path (the “curl”).

The Curling Ice Sheet

The size and preparation of the playing surface, or “curling sheet”, are specified by the rules of the World Curling Federation. It is a rectangular sheet of ice, as flat and level as possible, which is 45 m to 46 m long and 4.4 m to 5.0 m wide. The ice is typically refrigerated and its surface is maintained at a temperature of about -5°C. An essential part of the ice preparation is the spraying of water droplets onto the ice to form a “pebbled” surface. The tops of the pebbles are cut off, or “nipped”, with a blade. In a curling game, a stone is pushed, turned and released and then it travels, while rotating, about 25 to 30 m along the ice before coming to a stop. Typically, the lateral motion or curl is about 1 m.

The Curling Stone

The curling stone, also called a rock in North America, is essentially a stubby cylinder made of granite. Its dimensions are specified by the World Curling Federation. Its mass must be between 17 kg and 20 kg. The maximum allowed circumference is 910 mm and the minimum allowed height is 110 mm. The bottom of the curling stone is not flat. The outer perimeter is convex and does not touch the ice. The central region is concave and it too does not contact the ice. The only part of the stone in contact with the ice is the “running band”, which is a narrow, flat annulus, about 6.4 mm to 12.7 mm wide and 130 mm in diameter. The running band is typically made of blue hone granite, which has a low water absorptivity. This prevents repeated freeze/thaw cycles from eroding the running band. From time to time, the running band is “textured” or roughened, in order to maintain the curl. Using a jig, a curling technician slides the stone once forward over a distance of about 20 cm, on top of a sheet of medium grit, silicon carbide (P80) sandpaper. The stone is then rotated by about 120° and it is then pulled backwards to its starting point, across the same piece of paper. Typical particle diameters for this type of sandpaper range between 190 and 265 micrometres. The scratched surface of the running band is then wiped clean. At this point, it looks like Fig 1. A handle is attached to a bolt that runs vertically through a hole in the centre of the stone. The

handle allows the curler to grip and launch the stone and to rotate it upon release.

The Curling Broom

Typically, two team members travel down the ice, ahead of the stone, while rubbing the ice with a curling broom. Although the role of the curling broom can be critical to the strategy of a game by influencing a stone’s trajectory, it is not essential to our ice friction considerations and our attempt to ascertain why curling stones curl. Wheelchair curlers for example do not use curling brooms, and yet their curling stones still curl. Consequently, we shall not consider the effects of sweeping with a curling broom in this paper.



Fig. 1 The scratched surface of a curling stone running band (white band across the middle of the photograph). For scale, the Canadian dime is 18 mm in diameter.

Previous Scientific Work

Over the past century, the scientific investigation of curling has proceeded in fits and starts, but the rate of paper publication seems to have increased in the years leading up to the 2014 Winter Olympics in Sochi. The problem has occupied the attention of engineers, physicists, aerodynamicists and others. Bradley (2009) has published a recent review.

The earliest publication we have been able to find is that of Harrington (1924), working at the University of Saskatchewan. It reports on experiments on the motion and forces acting on curling stones, using both a tribometer and careful positional measurements made on an indoor curling rink. The author speculates that the cause of the curl is a differential frictional force on the left and right sides of the stone, so that the stone’s path curves towards the slow side of the stone (hereafter the H explanation). However, no quantitative model was proposed.

Unaware of the work of Harrington (1924), Macaulay and Smith (1930) at Cambridge presented an heuristic, analytical model of the curl that makes the assumption, attributed by them to Sir Gilbert Walker (hereafter the W explanation), that the friction at the back of the running band is greater than that at the front of the running band. Unfortunately, their simple model fails to predict the observed magnitude of the typical curl. A few months later, Harrington (1930) responded to Macaulay and Smith, arguing that the W explanation is not necessary and reiterating the H explanation. Thus began a

controversy that divided the scientific curling community into two camps – those who espoused the H explanation (sometimes referred to as left-right asymmetry) and those who espoused the W explanation (sometimes referred to as front-back asymmetry).

More than 50 years elapsed before another scientific study of curling stone dynamics appeared (Johnston, 1981). This paper was written by an aerodynamicist “for the entertainment and interest” of other aerodynamicists with an avocational interest in curling. It has no references and it uses aerodynamic terminology such as “pitching moment” and “drag, lift and moment coefficients”. Johnson espouses the W explanation. An analytical model is developed based on a “plausible, simple form for the contact pressure distribution”. However, no measurements are used as input to the model and the *assumed* friction coefficient is higher than observed. While the model produces some qualitatively reasonable predictions (for example, the trajectory geometry is insensitive to the initial angular velocity, as observed), there is no quantitative validation of the model.

Between 1996 and 2004, Mark Shegelski, his collaborators and students conducted an extensive series of theoretical and experimental investigations into the dynamics of curling stones (Shegelski et al., 1996; Shegelski and Niebergall, 1999; Shegelski et al., 1999; Shegelski 2000; Shegelski, 2001a; Shegelski, 2001b; Shegelski, 2002), culminating in the thorough and insightful paper of Jensen and Shegelski (2004). As a physicist, Shegelski espoused the W explanation. This led to a debate with Denny (Denny, 1998; Denny, 2002), who espoused the H explanation. The ensuing comments on each other’s papers (Shegelski and Reid, 2000; Denny, 2000; Denny, 2003; Shegelski and Niebergall, 2003) seem to have put an end to the H explanation, even in the popular literature on curling.

Penner (2001) presented theoretical and experimental results for sliding cylindrical shells and curling rocks. He states, “Left-right frictional asymmetry models [the H explanation] based on the conventional frictional model ... do not result in any net lateral force and therefore by themselves cannot explain the curl of a curling rock.” Nevertheless, he goes on to suggest that, if the friction is partly adhesive, it will result in preferential pivoting about the slow side of the rock. Although he concludes that this is “the primary mechanism behind the curl of a curling rock”, he was unable to formulate and test a quantitative model based on this mechanism.

Marmo and Blackford (2004) describe a numerical model of curling stone dynamics that is based on the friction coefficient derived by Stiffler (1984). It also includes the effect of sweeping on the stone’s trajectory. Unfortunately, the brevity of this paper did not permit the presentation of the full set of model equations, nor a complete validation of the results. The full model seems never to have been published, although additional information is given in Marmo et al. (2006) and Buckingham et al. (2006). The authors conclude: “The curved trajectory of rotating curling stones is due to a left-right frictional asymmetry caused by the dependence of friction upon the sliding velocity.” and “A numerical model based on equations of motion and an analytical solution for friction on melting surfaces correctly predicts the trajectory of slowly rotating curling stones.” Thus it appears that Marmo and Blackford have espoused the H explanation and produced a successful numerical model of curling stone dynamics. Unfortunately, too few model details have been published to allow an independent check of these results.

In the years leading up to the 2014 Winter Olympics in Sochi, no fewer than seven papers by four groups were published on curling ice friction and curling stone dynamics. Tusima (2010) addresses the problem

using his adhesion theory of ice friction. Ivanov and Shuvalov (2012 and 2013) show that isotropic friction models (friction coefficient constant around the perimeter) are able to explain the direction of the curl but not its magnitude. Anisotropic friction models can do both if the “correct” choice of model parameters is made. However, no attempt was made to derive the anisotropic model parameters based on first principles. Maeno (2013) presents an attempt to develop a model of curling stone dynamics, based on an evaporation-abrasion model of ice friction. In deriving the model, he introduces two new parameters, the “friction magnification factor” (Maeno, 2010 and 2013) and the “curl ratio”. However, it is not entirely clear how to derive these parameters from first principles. Nyberg and his colleagues published two papers on curling stone dynamics (Nyberg et al., 2013a; Nyberg et al., 2013b). The first paper purports to show that “no anisotropic friction mechanism can explain the observed motion of a real curling stone.” The authors conclude that it is still challenging to answer the question, “What puts the curl in a curling stone?” Unfortunately, these authors appear to be unaware of the results of Ivanov and Shuvalov (2013), who present an anisotropic friction model that does so. In the second paper, the authors present a theory of scratch-guided motion, but it has not yet been quantified or validated. Shegelski is quite skeptical about this proposed scratch mechanism (Shegelski et al., 2015a; Shegelski et al., 2015b), because it makes no quantitative predictions (there are no equations) and it fails a number of simple tests when one compares its qualitative predictions with observations and measurements of curling stone behavior.

Thus, despite almost a century of analysis and experiments on curling stone dynamics, the words of Macaulay and Smith written in 1930 still resonate: “It seems clear that we have failed to take account of some important feature of the motion. What is it?” At the time of writing, there is no numerical model of curling stone dynamics that satisfactorily predicts all of the observations, both qualitatively and quantitatively. Nevertheless, one should not be too pessimistic. The “wet friction” models of Shegelski and his colleagues (e.g. Jensen and Shegelski, 2004) explain a number of the observed aspects of curling stone dynamics. One should not dismiss them merely because they do not fully explain all of the observations.

OBSERVATIONS

Here we enumerate those qualitative and quantitative aspects of curling stone dynamics that will need to be explained by any successful numerical model. They are organized into three categories – no curl, “normal” curl and enhanced curl. But first, we begin with a few general aspects of the behavior of curling stones that also need to be reproduced by any successful model (Jensen and Shegelski, 2004).

- 1.1 The trajectory of a rotating-translating curling rock is not a straight line. Instead, it is curved (some call it “J-shaped”).
- 1.2 The direction of the lateral motion, the “curl”, is in the same sense as the rotational motion of the front of the stone. That is, if the front of the stone moves towards the right, then the stone curls towards the right and vice versa. This is the opposite of what happens if one slides a cylinder with a similar ring contact over a smooth, dry surface.
- 1.3 The lateral displacement or “curl” is typically about 1 m in the game of curling. We shall refer to this value as “normal” curl.
- 1.4 The curl is almost independent of the rotational speed imparted to the stone, over a range of total number of rotations (total turns) from about 1 to 25 (Jensen and Shegelski, 2004; Penner, 2001).
- 1.5 In standard curling play, the stone’s translational and rotational motion stop at the same time. For rapidly rotating and slowly sliding stones, rotation can continue after translation stops (Shegelski and Niebergall, 1999).

1.6 Sweeping helps a stone to travel farther and straighter (less curl).

Jensen and Shegelski (2004) also present numerous detailed experimental results, summarized in their Figs. 2, 3, 5, 7 and 8. These data could be used to test new quantitative predictive models. We will do so later in this paper.

No Curl

There are several circumstances when a curling stone will not curl or will exhibit very little curl. Unless we specify otherwise, we are speaking of a “normal” curling stone (one that could be used in Olympic curling for example) that is thrown “normally” and travels over “normally” pebbled ice.

2.1 A curling stone does not curl if it is thrown without rotation (translation without rotation). However, a stone thrown without initial rotation can begin to rotate and curl spontaneously. For a total number of rotations between 0 and 1, the curl increases approximately linearly with number of rotations (Penner, 2001).

2.2 A curling stone does not curl if it is set into pure rotational motion (rotation without translation).

2.3 A curling stone does not curl if its running band has been polished so that it lacks the normal texturing (Nyberg et al., 2013a).

2.4 A curling stone does not curl if it is thrown on burned ice, an ice surface that has been scraped, then melted and refrozen, leaving a glassy smooth surface without significant roughness (Jensen and Shegelski, 2004).

Normal Curl

3.1 The essential requirements for normal curl are translation and rotation of the curling stone. In addition, the ice must have some roughness, as must the running band of the stone.

Enhanced Curl

4.1 Curling rocks thrown on scraped ice exhibit enhanced but uncontrolled curl. Scraped ice is the term used to describe the ice surface following mechanical scraping of pebbled ice with a lower blade than that used for nipping, so that all of the pebble is removed. Scraping is performed routinely in order to resurface a curling sheet. It leaves an ice surface with considerable roughness over a wide range of length scales.

4.2 Curling rocks thrown on flooded ice exhibit enhanced curl (up to ten times the lateral acceleration of normal curling). Flooded ice is the term used to describe the ice surface after flooding the rink and allowing the water to freeze. It typically has considerable, small-scale roughness, but it is not as durable as pebbled ice.

4.3 Curling rocks thrown on hockey ice can exhibit extremely large curls (Shegelski and Holenstein, 2002).

4.4 Curling rocks thrown with a rotational speed producing a total number of rotations (total turns) between about 25 and 75 exhibit enhanced curl that increases more or less linearly with the total turns (Jensen and Shegelski, 2004).

4.5 Curling rocks with a larger diameter running band exhibit enhanced curl (Shegelski, 2001b).

4.6 Shegelski devised a special curling stone with a polished, stainless steel running band. Unlike 2.3, this stone exhibited enhanced curl when thrown on a standard, pebbled ice sheet.

MODEL OF ICE PEBBLE-STONE FRICTION

In order to estimate the overall forces and torques that determine a

curling stone’s trajectory, we shall begin by considering the forces that arise as a result of the interaction between the running band and a single ice pebble.

Contact Geometry and Ploughing Force

We consider here the lubricated contact between curling ice and the running band. Both are rough surfaces and the contact details will be determined by the roughness scale on each surface. On curling ice, the dominant roughness scale is that of the ice pebbles. Individual pebbles are of the order of 1 mm high and a few millimetres in equivalent diameter, namely the diameter of a circle with the same area as that of the top of a nipped pebble (Fig. 2). Mean pebble spacing is of the order of 5 to 10 mm. The freshly nipped pebbles look like truncated circular cones. Some pebbles in Fig. 2 are not high enough to have been cut by the nipper.

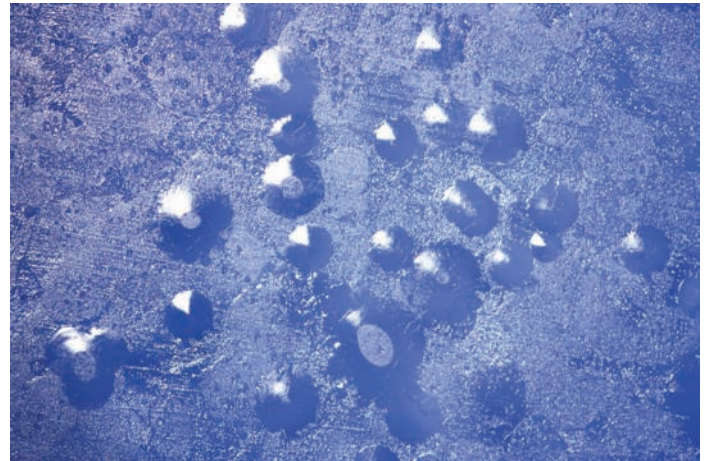


Fig. 2 Pebbles on freshly nipped ice at the Calgary Curling Club. The appearance of the nipped pebbles is that of truncated, circular cones. Same scale as in Fig 1.

Nyberg et al (2013a) have measured the surface topography of pebbles. Freshly prepared (nipped) and swept pebbles have roughness heights on their top surfaces of the order of 1 micrometre or less. Consequently, for our purposes, we shall consider the tops of the pebbles to be perfectly smooth. We shall also assume that they are flat. These authors have also measured the surface topography of a scratched running band. Roughness height is of the order of 1 to 10 micrometres, while its horizontal scale varies from about 50 to 500 micrometres. Given that ice hardness is an order of magnitude less than that of granite (and that ice can readily melt), it would seem reasonable that the contact between the running band and a pebble is controlled by the roughness of the running band.

The apparent contact area between the pebble and the running band is simply the area of the top of the pebble, A_p . In order to determine the actual contact area, we shall make the same assumption as Makkonen and Tikanmaki (2014) and Lozowski et al. (2013), namely that there is a vertical force balance such that the product of the ice hardness and actual contact area equals the applied load. When the weight of the stone is supported by multiple pebbles, it is not apparent how the weight is distributed among the individual pebbles. Moreover, because of stone deceleration, the load on the front will be somewhat higher than the load on the back. For simplicity, we will ignore both effects and assume that the weight of the stone is divided equally among all of the contacting pebbles.

The penetration depth into the ice of the roughness elements on the running band can be estimated as follows. Consider the apparent contact area *on the running band*, with a total area equal to the top surface area of the pebble, A_p . Let $f(d)$ be the fractional area (projected onto the top surface of the pebble), with a depth d below a reference level. The reference level is the level above which there are no air gaps in the granite due to roughness. Then, according to our vertical force balance assumption:

$$f(d)A_p p(T) = L_p \quad (1)$$

where $p(T)$ is the ice hardness (a function of ice surface temperature T), and L_p is the load on the pebble (the portion of the weight of the stone acting on the pebble).

If the function f can be determined experimentally, or assumed via modelling, then Eq. 1 can, in principle, be used to determine d , since all of the other quantities in the equation are known.

In the absence of measurements of the function f for an actual running band, let us consider a simple roughness model. We will assume that all asperities are identical circular cones of aspect ratio α (height divided by radius). The load, L_a , supported by each asperity is given by:

$$L_a = \frac{mg}{n_p n_a} \quad (2)$$

where n_p is the number of pebbles supporting the stone and n_a is the number of asperity contacts per pebble.

The vertical force balance for a single asperity is given by:

$$L_a = \pi r_c^2 p \quad (3)$$

where p is the ice hardness and r_c is the contact radius, at the ice surface, between the asperity and the ice. This equation can be solved for r_c and the contact depth, $h_c = r_c \alpha$, can then be calculated.

The ploughing force per asperity is given by:

$$F_p = p h_c r_c = p r_c^2 \alpha = L_a \alpha / \pi \quad (4)$$

Hence the friction coefficient due to ploughing is given by:

$$\mu_p = \frac{F_p}{L_a} = \frac{\alpha}{\pi} \quad (5)$$

We note that this value is independent of the load, the number of contacting pebbles and the number of asperities per pebble.

If we assume that the asperities are identical and support identical loads, the friction coefficient due to ploughing for an ensemble of asperities is also given by α/π . For an asperity aspect ratio of .01, the friction coefficient due to ploughing is 3.2×10^{-3} . In order to obtain a precise value, it is imperative to measure f for the running band, or at least to find a means to estimate the aspect ratio of the roughness elements. Nevertheless, we shall discover a way to bypass this requirement in what follows.

At this point, we assume the following expression for ice hardness, based on measurements made on various athletic ice surfaces (Poirier et

al., 2011):

$$p = 14.7 - 0.6T \quad (6)$$

Lubrication Equation and Shear Stress Force

Friction between the running band of the stone and an ice pebble produces heat, which melts ice at the interface and gives rise to a lubricating film of liquid water (Shegelski et al, 1999; Shegelski and Niebergall, 2003). We have made a similar assumption in our previous work on ice friction in speed skating (Lozowski et al., 2013) and we shall make it again here. Makkonen and Tikanmaki (2014) suggest that lubricated friction is a reasonable assumption for sliding speeds in excess of about 10 cm s^{-1} , or for all but the very last part of a stone's trajectory.

We shall adopt the approach of Lozowski et al. (2013) and so we will not repeat the detailed derivation of the lubrication equation here. In that work, we ignored heat conduction into the slider and the heat generated by the ploughing force, but we will consider them here. Moreover, because the ice/slider contact in curling is not elongated, as in skating, we shall consider the development of the lubrication layer in time rather than in space. Hence, mutatis mutandis, the lubrication equation for the contact between a pebble and the running band of the stone is given by:

$$\frac{dh}{dt} = \frac{\alpha p V}{\pi \rho_w l_f} + \frac{\mu V^2}{h \rho_w l_f} - \frac{k_i \Delta T_i}{\sqrt{\pi} \kappa_i t_p \rho_w l_f} - 2 \frac{k_g \Delta T_g}{\sqrt{\pi} \kappa_g t_p \rho_w l_f} - \frac{2 m g h^3}{3 \pi \mu n_p r_p^4} \quad (7)$$

where the variables and parameters are defined below. Units, ranges and standard values are indicated in square brackets.

h	lubrication layer thickness [m]
t	time [s]
μ	dynamic viscosity of water [$1.79 \times 10^{-3} \text{ Pa s}$]
V	stone translational speed [0.1 to 4 m s ⁻¹]
ρ_w	density of liquid water [$1.0 \times 10^3 \text{ kg m}^{-3}$]
l_f	specific latent heat of freezing of water [$3.5 \times 10^5 \text{ J kg}^{-1}$]
k_i	thermal conductivity of ice [$2.3 \text{ W m}^{-1} \text{ K}^{-1}$]
k_g	thermal conductivity of Blue Hone granite [$2.94 \text{ W m}^{-1} \text{ K}^{-1}$]
ΔT_i	absolute value of difference between ice temperature and freezing point temperature based on $p(T)$
ΔT_g	absolute value of difference between running band temperature and freezing point temperature based on $p(T)$
κ_i	thermal diffusivity of ice [$1.23 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$]
κ_g	thermal diffusivity of Blue Hone granite [$1.49 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$]
$t_p = 2r_p/V$	
m	stone mass [17 to 20 kg]
g	acceleration of gravity [9.8 m s^{-2}]
n_p	number of pebbles contacting running band [10 to 100]
r	inner radius of the running band [60 mm]
r_p	radius of top contact of pebble [0.5×10^{-3} to $2.5 \times 10^{-3} \text{ m}$]
w	width of running band [6.4 to 12.7 mm, standard value 9 mm]

In Eq. 7, the left-hand-side is the rate of change of the thickness of the lubrication melt layer on the top of the pebble. The first term on the right arises from the work done by the ploughing force. The second term arises from the work done by the shear stress in the lubrication layer. The third term arises from heat conduction into the ice. The fourth term arises from heat conduction into the running band. It differs

from the third term because we have averaged over the time of contact between the pebble and a point on the running band. (There is an asymmetry in the heat conduction terms, because the contact time between the running band and a point on the pebble, t_b , differs from the contact time between the pebble and a point on the running band, t_p .) The fifth term on the right arises from the squeeze flow of liquid out of the contact between the pebble and the running band.

In principle, Eq. 7 can be solved to determine $h(t)$, which can then be used to determine the time dependent shear stress force. This can then be added to the constant ploughing force to determine the total resistive force and hence the coefficient of friction. Instead of doing this, we shall first consider a simpler approach below. This energy balance approach does not actually presuppose any knowledge of the friction mechanism. The basic assumptions are that friction produces heat, the heat is sufficient to warm the surface of the ice to 0°C , and all of the energy dissipated by friction is conducted into the ice and the running band. Our derivation will begin with Eq. 7. However, it is possible to skip Eqs. 7, 8 and 9 and to write Eq. 10 directly, based on the three assumptions given above.

Energy Balance Approach

We will assume that an equilibrium lubrication layer thickness is reached quickly in the solution to Eq. 7, and hence $dh/dt = 0$. We will also assume that the lubrication layer is sufficiently thin that the squeeze flow term, which depends on h^3 , is negligible. Having made these assumptions, let us consider a balance of the energy fluxes produced by friction (including the heat produced by the ploughing force and the shear stress) and the energy fluxes by conduction into the stone and ice. Both conduction heat fluxes are now averaged over the appropriate time intervals. For heat conduction into the stone, this is the time required for a point on the running band to pass across the diameter of the pebble. For heat conduction into the ice, it is the time required for a point on the pebble to pass under the running band. We assume that, in between contact with individual pebbles, the running band cools down to the same initial temperature that it had when the stone was sitting on the ice. Hence, the energy balance can be written:

$$\frac{\mu V^2}{h} + \frac{\alpha p V}{\pi} = \frac{2k_g \Delta T_g}{\sqrt{\pi \kappa_g t_p}} + \frac{2k_i \Delta T_i}{\sqrt{\pi \kappa_i t_b}} \quad (8)$$

where t_b ranges between $t_b = w/V$ (which we shall call t_{bmin}) and $\frac{2(2rw + w^2)^{1/2}}{V}$ (which we shall call t_{bmax}).

The left side of Eq. 8 is the total heat flux generated by the total friction force (the sum of the shear stress force and the ploughing force). Hence it may be written:

$$\frac{\mu V^2}{h} + \frac{\alpha p V}{\pi} = \mu_t p V \quad (9)$$

where μ_t is the total friction coefficient. Eq. 9 is similar to that derived by Stiffler (1984) and used by Marmo et al (2006), except that they both ignored heat conduction into the ice. Equating the right-hand-sides of Eqs. 8 and 9, the total friction coefficient can be solved for:

$$\mu_t = \frac{2}{pV} \left[\frac{k_g \Delta T_g}{\sqrt{\pi \kappa_g t_p}} + \frac{k_i \Delta T_i}{\sqrt{\pi \kappa_i t_b}} \right] \quad (10)$$

where p is the ice hardness measured by Barnes and Tabor (1968), and parameterized by Makkonen and Tikanmaki (2014) [We have corrected a sign error in Makkonen and Tikanmaki]. While the solution given by Eq. 10 is convenient and simpler than solving Eq. 7, it does have some drawbacks. By reducing the calculation of friction to a thermodynamic balance, it bypasses the role of running band roughness. And yet periodic roughening or texturing of the running band is an essential aspect of the actual game. Eq. 10 also bypasses an explicit consideration of the lubrication layer and its thickness. While layers of water exceeding about ten molecules in thickness have properties similar to those of bulk water (Israelachvili et al., 1988), if the melt water layer is thinner than this, the parameters on the right-hand-side of Eq. 9 may no longer be valid. For now, however, we shall ignore these drawbacks and proceed to use this solution. Fig. 3 shows two solutions of Eq. 10, using t_{bmin} and t_{bmax} . We have also assumed that $\Delta T_g = \Delta T_i$ and that both are equal to 5 K (Marmo and Blackford, 2004). The total friction coefficient is clearly a decreasing function of running band speed, decreasing rapidly at speeds below about 0.5 ms^{-1} and more slowly thereafter. The curves are broadly consistent with the friction measurements of others for a curling stone, although a bit low for speeds exceeding 1 ms^{-1} , and much higher than for the polished running band at low speeds.

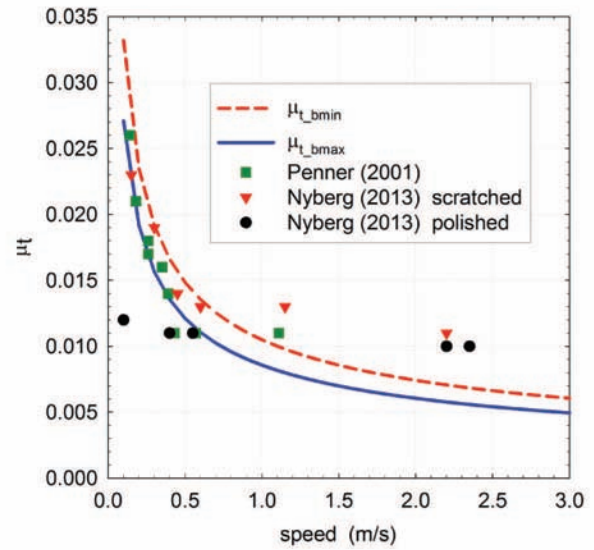


Fig. 3 Total friction coefficient (curves) for a single pebble as a function of running band speed, over the range of possible contact times between the pebble and the running band, in purely translational motion. Points are experimental measurements for a rock.

Case Study: Translation without Rotation

We can now use this friction coefficient, *predicted from first principles*, to solve the equations of motion for a curling stone in purely translational motion (no rotation). Purely rotational motion is considered in Appendix A. In order to predict a stone's trajectory, it is important to understand that the friction coefficient for the entire stone is the same as that for a single pebble. This is because, under our assumption of identical pebbles which support identical loads, the total load for the stone is n_p times higher than that for a single pebble, but so is the total friction force. So their ratio remains constant. Hence the equations of motion may be written:

$$\frac{dV}{dt} = -g\mu_t(V) \quad (11)$$

$$\frac{dy}{dt} = v \quad (12)$$

where we use y to denote the longitudinal coordinate, following the usage of Jensen and Shegelski (2004).

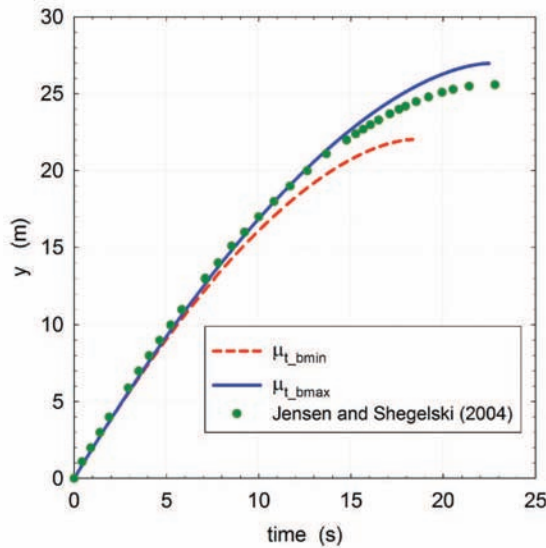


Fig. 4 Trajectory of a curling stone in purely translational motion, for contact times between the running band and individual pebbles spanning the range between the maximum and minimum values. Green points are experimental data (Jensen and Shegelski, 2004; their Fig. 5).

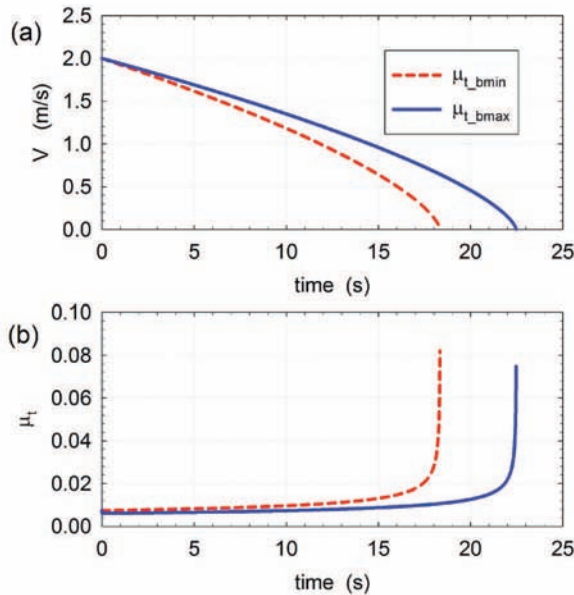


Fig. 5 Curling stone speed (a) and friction coefficient (b) as functions of time.

Fig. 4 shows the solution for a stone with an initial velocity of 2 ms^{-1} . The resulting longitudinal trajectories resemble those measured by Jensen and Shegelski (2004) for a normally played, rotating curling stone (a so-called “draw shot”). We have compared the model with a rotating stone, because there are no comparable trajectory data for non-rotating stones. This is, in part, because non-rotating, translating stones

often start to rotate spontaneously (Daniel Bubola, pers. comm.). Fig. 5 shows the curling stone speed as a function of time. The slope is the acceleration, which is proportional to the friction coefficient (Eq. 11). Over that last few seconds of the trajectory, the friction coefficient increases dramatically. For the blue trajectory, the stone takes almost 20 seconds to slow from 2 ms^{-1} to 0.5 ms^{-1} . But thereafter, it takes only 3 seconds to stop. Fig. 6 shows the stopping distance of the stone as a function of its initial speed. If the friction coefficient were independent of speed, this relationship would be quadratic. However, it is apparent that this is not the case in Fig. 6. For example, the stopping distance when launched at 1.0 ms^{-1} is about 5 m (blue curve). A simple quadratic relation would predict a stopping distance of 25 m when launched at 2.0 ms^{-1} , but the actual trajectory stops at 27 m.

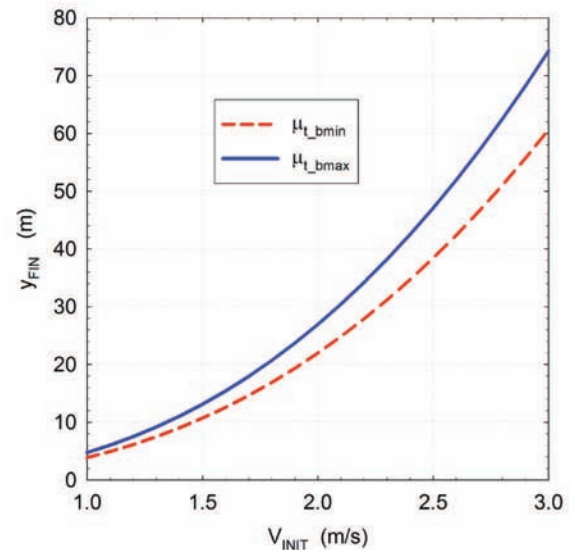


Fig. 6 Stopping distance vs initial stone speed.

DISCUSSION AND CONCLUSIONS

By dissecting and simplifying the problem of ice friction in curling, we have made some progress in formulating the ice friction coefficient for a curling stone on pebbled ice using a thermodynamic approach. Starting from first principles, we have derived an equation for the total ice friction coefficient for a single pebble. It is in broad agreement with measurements of the ice friction coefficient for a stone. We have used the model to predict trajectories for a translating but non-rotating curling stone and also for a rotating but non-translating stone. A comparison with our own experimental data and with experimental data of Jensen and Shegelski (2004) is very encouraging, even though the Jensen and Shegelski data are for a rotating stone. We have used the theory to predict the stopping distance of a curling stone as a function of its initial speed. This graph could be used to test our model. In principle, it could also be used by curlers to improve their game, although the best curlers probably already have such a mental function in their head. The stopping distance is very sensitive to initial velocity. Only a 1% change in the initial velocity can lead, without sweeping, to almost a 1 m change in the stopping distance. Hence, to play the game well, even longitudinally, requires considerable expertise, especially for wheelchair curlers. We have not attempted to explain the curl using a friction force directed antiparallel to the local velocity of the running band relative to the ice. Research to explain the curl quantitatively is ongoing.

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APPENDIX A

Case Study: Rotation without Translation

In another paper, we will consider the trajectories of real curling stones, with coupled translation and rotation. However, before doing so, we will consider the complementary case to translation without rotation, namely rotation without translation. With minor modifications, it is easy to model using Eq. 10. Moreover, purely rotating experiments are easy to perform and we have done so.

The rotational equations of motion are:

$$\frac{d\theta}{dt} = \omega \quad (13)$$

$$T = I \frac{d\omega}{dt} \quad (14)$$

where θ is the angle of rotation around the centre of mass, ω is the angular velocity of the stone around the centre of mass, T is the frictional torque exerted at the ice pebble/running band interface, and I is the moment of inertia of the stone, given by:

$$I = \frac{mR^2}{2} \quad (15)$$

where m is the mass of the stone, and $R = 0.145m$ is the radius of the stone (not the radius of the running band).

The torque is given by:

$$T = -\mu_t(t)mgr \quad (16)$$

where r is the mean radius of the running band.

The friction coefficient is time dependent because of deceleration. Time dependence also arises because t_b in Eq. 10 must be replaced by t (the actual elapsed time since the rotation was started), because we assume that friction keeps the top of each pebble at the freezing point for the entire duration of the spin.

Hence:

$$\mu_t(t) = \frac{2}{p\omega(t)r} \left[\frac{k_g \Delta T_g}{\sqrt{\pi \kappa_g t_p}} + \frac{1}{2} \frac{k_i \Delta T_i}{\sqrt{\pi \kappa_i t}} \right] \quad (17)$$

Eq. 17 also differs from Eq. 10 because V has been replaced by ωr and t_p becomes:

$$t_p = \frac{2r_p}{\omega r} \quad (18)$$

Fig. 7 shows the numerical solution of Eqs. 13 to 18, with an initial angular velocity of 10 radians s^{-1} . The blue dots are experimental points, measured from a video we made at the Calgary Curling Club. A curling stone was rotated by hand back and forth a couple of times on the ice. Then without stopping, it was given an initial angular velocity of approximately 10 radians s^{-1} and left to spin down. The entire process was recorded from above using 30 fps HD video. The video was used to extract the timing points after every full rotation.

The agreement between the theoretical curve and the experimental points is quite good, although the numerical curling stone stops sooner than the actual stone. One possible explanation for this difference is that we started the clock at the time the stone was released to rotate freely. The initial hand rotation in the experiment may have "primed" the stone with a lower initial coefficient of friction than the numerical stone. More experiments would help to elucidate whether or not there is a systematic difference.

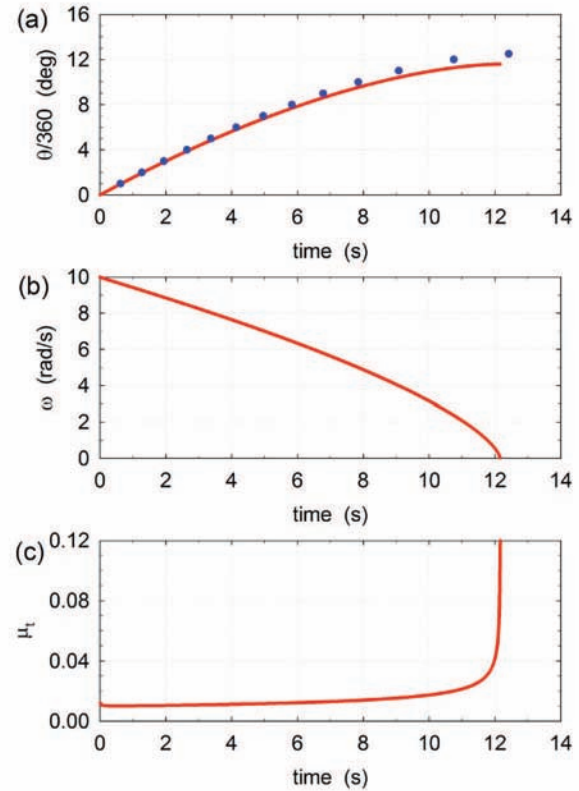


Fig. 7 (a) angular velocity, (b) rotation angle, (c) friction coefficient for pure rotation of a curling stone. The red curves are the numerical model solutions. The blue points are measurements made by us.

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